FCUSB

Перв. примен.

Справ. №

Дата

ПЗ:

Подп. и дата

Инв. № дубл.

Взам. инв. №

Подп. и дата

Инв. № подл.

Изм	Лист	№ докум.	Подп.	Дата
Разраб.		Аникин		
Пров.		Тарасов		
Т.контр.				
Н.контр.				
Утв.		*		

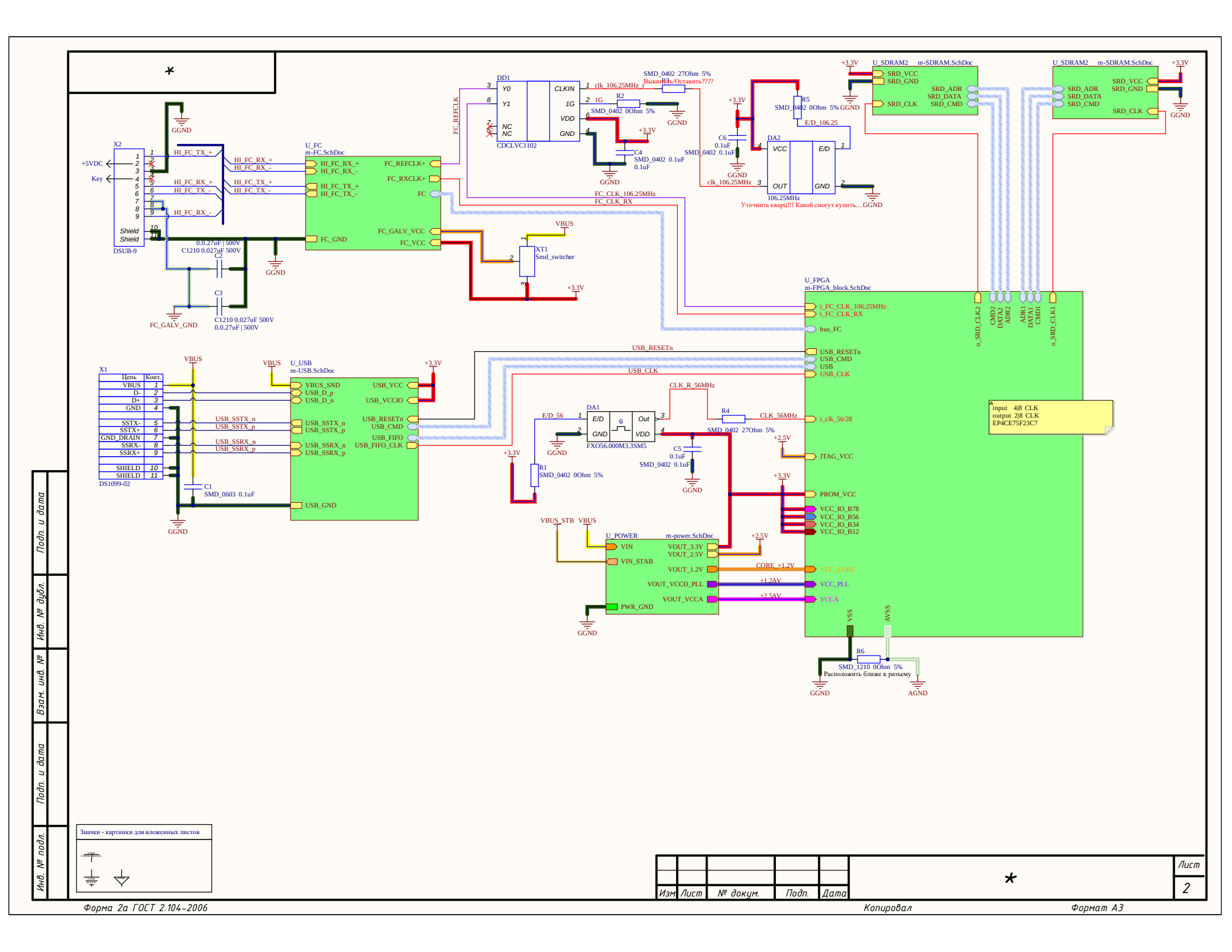
FCUSB

Литера	Масса	Масштаб

Лист 1	Листов 13

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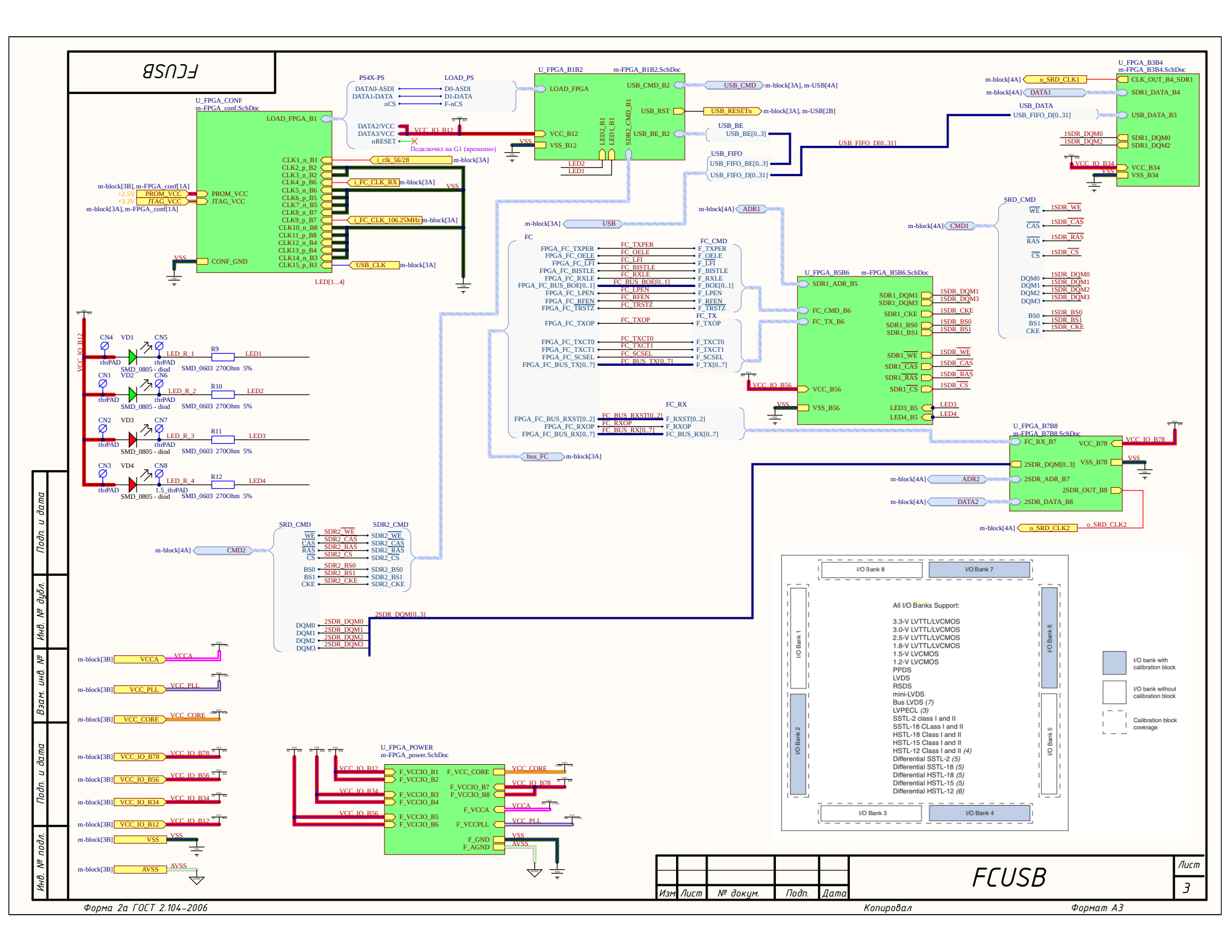
Форма 1 ГОСТ 2.104-2006 Копировал Формат А4

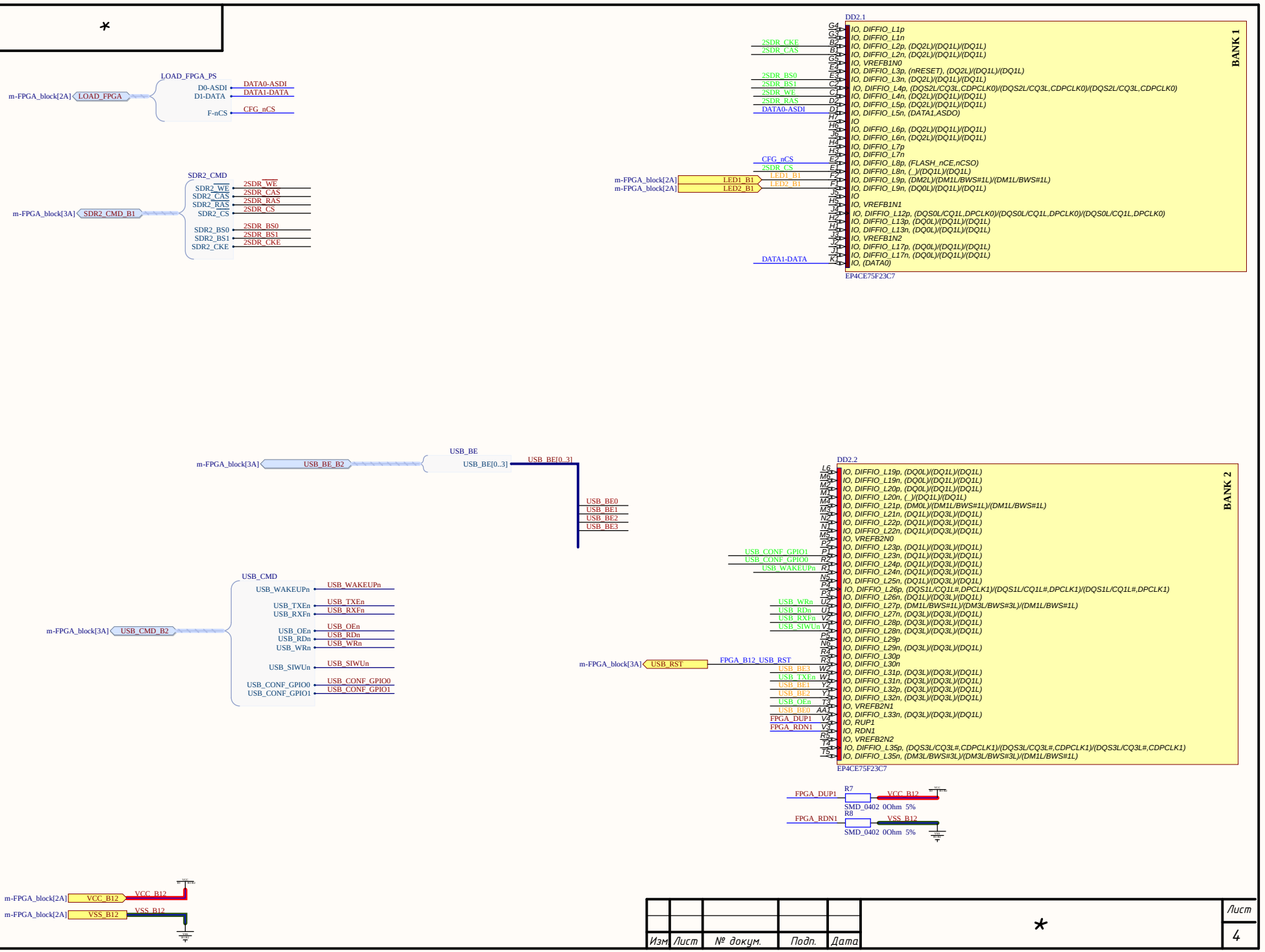


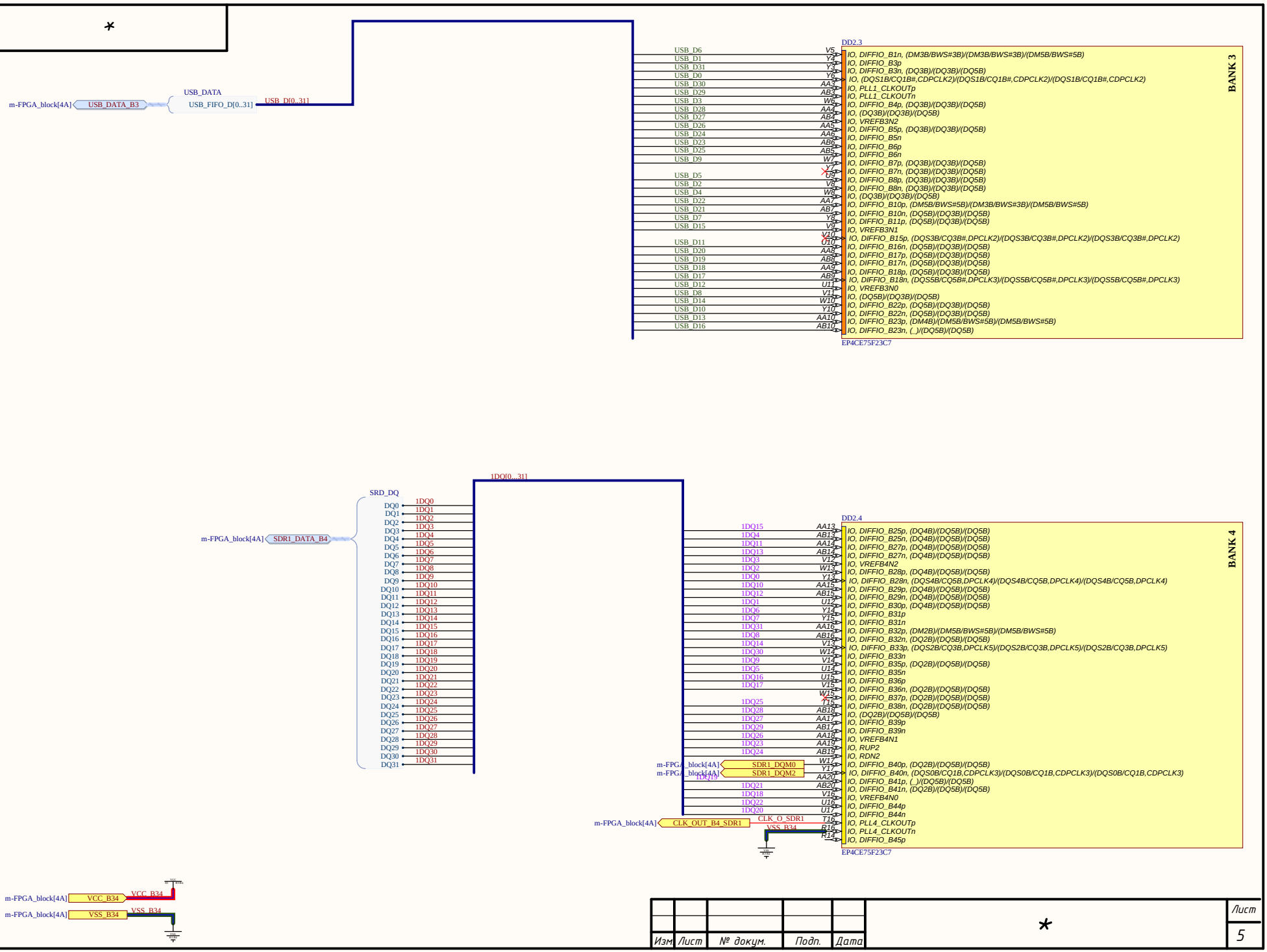
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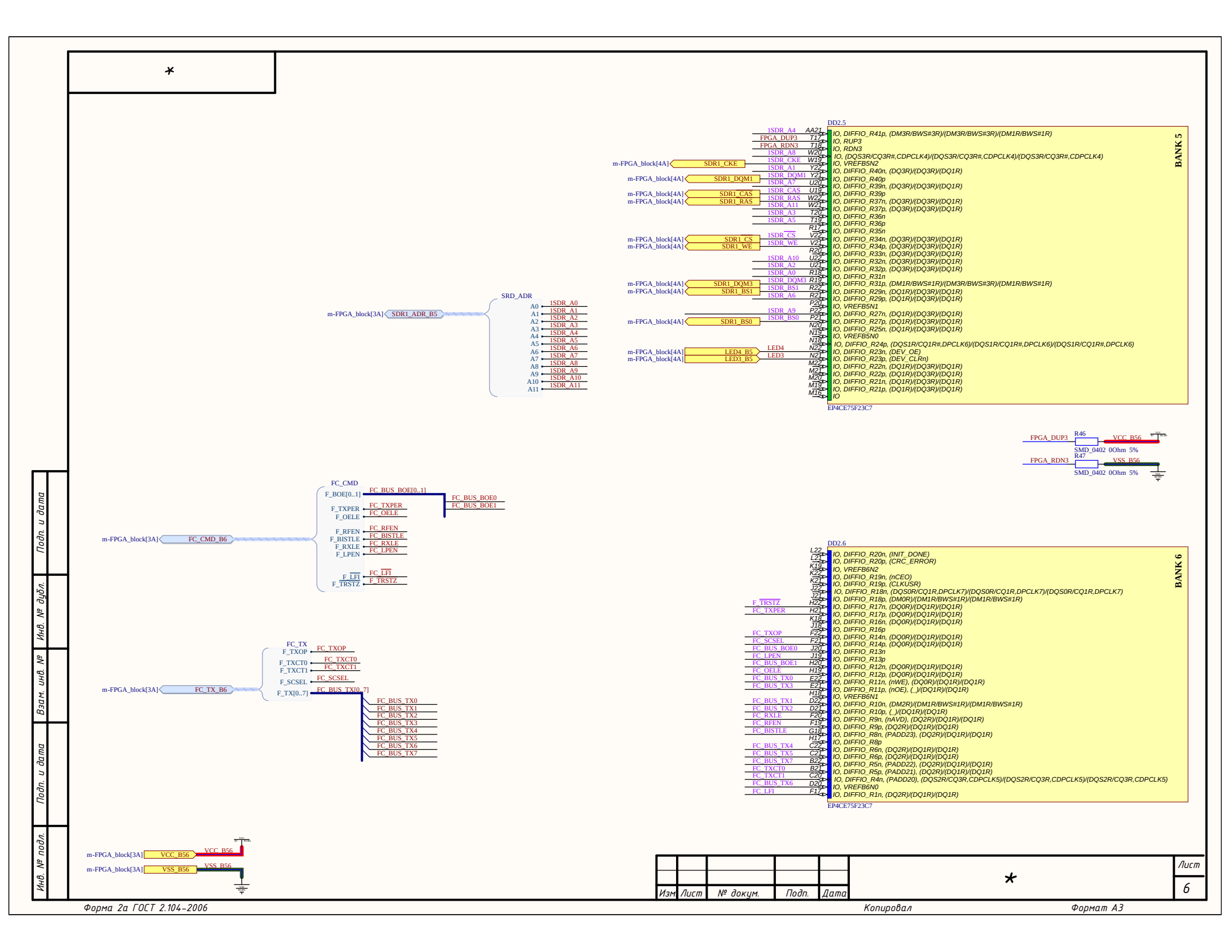
Изм	Лист	№ докум.	Подп.	Дата	*	Лист 2

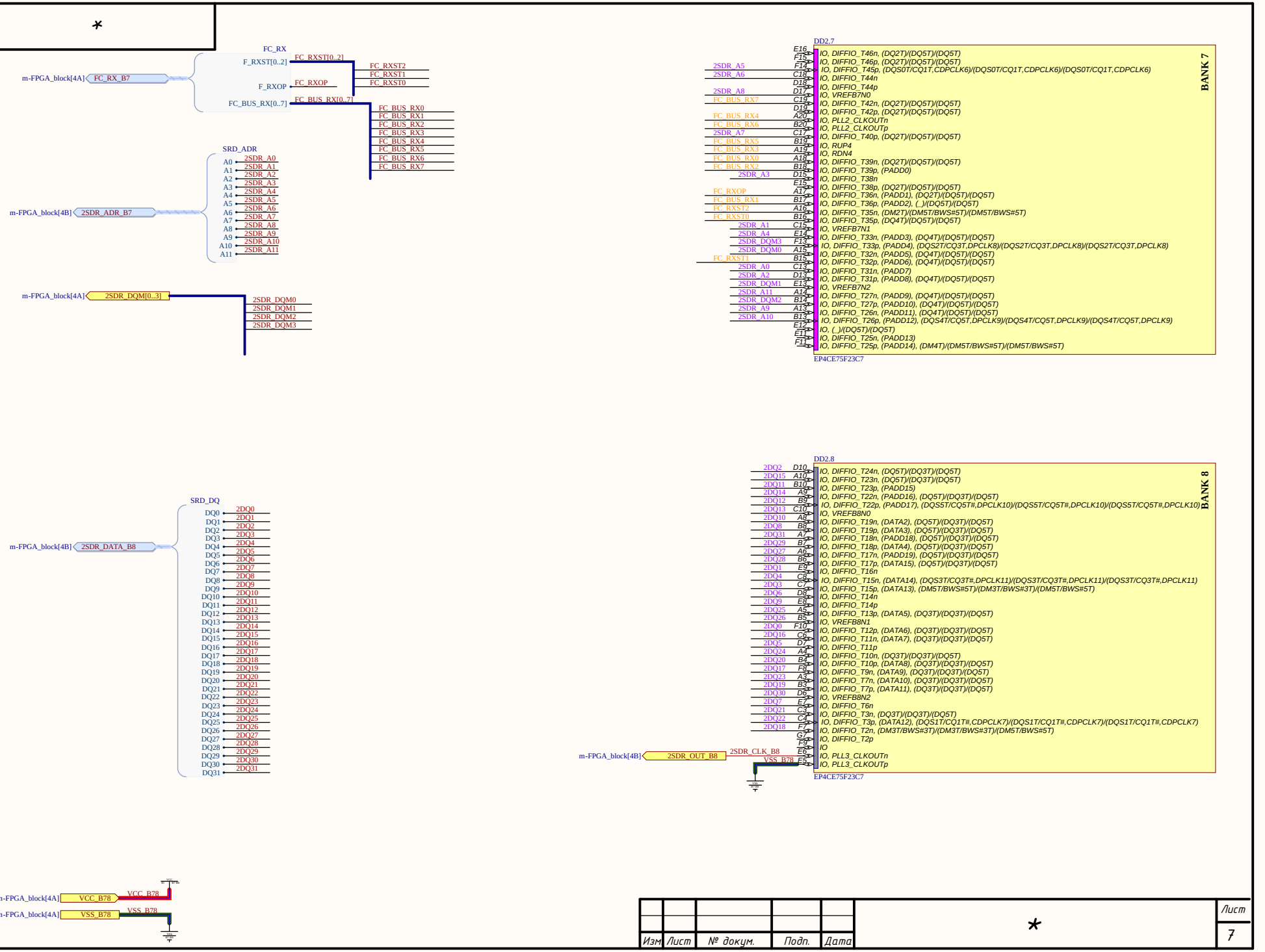
Форма 2а ГОСТ 2.104-2006 Копировал Формат А3

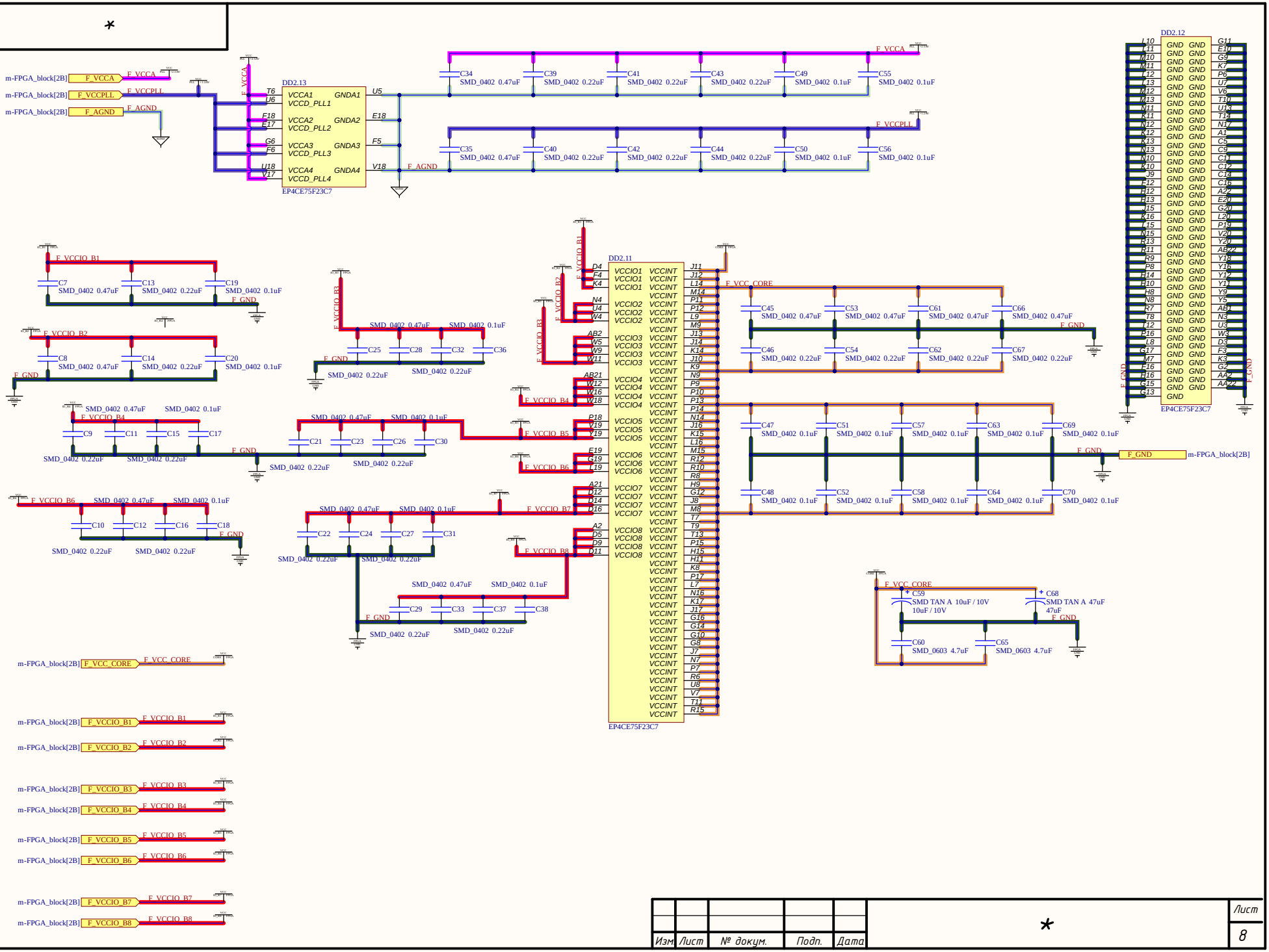

FCUSB
U_FPGA_CONF
m-FPGA_conf.SchDoc
LOAD_FPGA_B1
PROM_VCC
JTAG_VCC
CONF_GND
U_FPGA_B1B2
m-FPGA_B1B2.SchDoc
LOAD_FPGA
VCC_B12
VSS_B12
USB_CMD_B2
USB_RST
USB_BE_B2
U_FPGA_B3B4
m-FPGA_B3B4.SchDoc
CLK_OUT_B4_SDR1
SDR1_DATA_B4
USB_DATA_B3
SDR1_DQM0
SDR1_DQM2
VCC_B34
VSS_B34
U_FPGA_B5B6
m-FPGA_B5B6.SchDoc
SDR1_ADR_B5
FC_CMD_B6
FC_TX_B6
VCC_B56
VSS_B56
U_FPGA_B7B8
m-FPGA_B7B8.SchDoc
FC_RX_B7
2SDR_DQM[0..3]
2SDR_ADR_B7
2SDR_OUT_B8
2SDR_DATA_B8
VCC_B78
VSS_B78
U_FPGA_POWER
m-FPGA_power.SchDoc
F_VCCIO_B1
F_VCCIO_B2
F_VCCIO_B3
F_VCCIO_B4
F_VCCIO_B5
F_VCCIO_B6
F_VCCIO_B7
F_VCCIO_B8
F_VCC_CORE
F_VCCA
F_VCCPLL
F_GND
F_AGND
PS4X-PS
LOAD_PS
DATA0-ASDI
DATA1-DATA
nCS
D0-ASDI
D1-DATA
F-nCS
DATA2/VCC
DATA3/VCC
nRESET
Подключил на G1 (временно)
USB_CMD
USB_RESETn
USB_BE
USB_BE[0..3]
USB_FIFO
USB_FIFO_BE[0..3]
USB_FIFO_D[0..31]
USB_DATA
USB_FIFO_D[0..31]
i_clk_56/28
i_FC_CLK_RX
i_FC_CLK_106.25MHz
USB_CLK
LED[1...4]
LED1
LED2
LED3
LED4
FC
FC_CMD
FC_TX
FC_RX
FPGA_FC_TXPER
FPGA_FC_OELE
FPGA_FC_LFI
FPGA_FC_BISTLE
FPGA_FC_RXLE
FPGA_FC_BUS_BOE[0..1]
FPGA_FC_LPEN
FPGA_FC_RFEN
FPGA_FC_TRSTZ
FPGA_FC_TXOP
FPGA_FC_TXCT0
FPGA_FC_TXCT1
FPGA_FC_SCSEL
FPGA_FC_BUS_TX[0..7]
FPGA_FC_BUS_RXST[0..2]
FPGA_FC_RXOP
FPGA_FC_BUS_RX[0..7]
SRD_CMD
SDR2_CMD
SDR1_DQM1
SDR1_DQM3
SDR1_CKE
SDR1_BS0
SDR1_BS1
SDR1_WE
SDR1_CAS
SDR1_RAS
SDR1_CS
LED_R_1
LED_R_2
LED_R_3
LED_R_4
SMD_0805 - diod
SMD_0603 270Ohm 5%
VCCA
VCC_PLL
VCC_CORE
VCC_IO_B78
VCC_IO_B56
VCC_IO_B34
VCC_IO_B12
VSS
AVSS
I/O Bank 1
I/O Bank 2
I/O Bank 3
I/O Bank 4
I/O Bank 5
I/O Bank 6
I/O Bank 7
I/O Bank 8
All I/O Banks Support:
3.3-V LVTTL/LVCMOS
3.0-V LVTTL/LVCMOS
2.5-V LVTTL/LVCMOS
1.8-V LVTTL/LVCMOS
1.5-V LVCMOS
1.2-V LVCMOS
PPDS
LVDS
RSDS
mini-LVDS
Bus LVDS (7)
LVPECL (3)
SSTL-2 class I and II
SSTL-18 CLass I and II
HSTL-18 Class I and II
HSTL-15 Class I and II
HSTL-12 Class I and II (4)
Differential SSTL-2 (5)
Differential SSTL-18 (5)
Differential HSTL-18 (5)
Differential HSTL-15 (5)
Differential HSTL-12 (6)
I/O bank with calibration block
I/O bank without calibration block
Calibration block coverage
Подп. и дата
Инв. № дубл.
Взам. инв. №
Подп. и дата
Инв. № подл.
Изм
Лист
№ докум.
Подп.
Дата
FCUSB
Лист
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Форма 2а ГОСТ 2.104-2006
Копировал
Формат А3

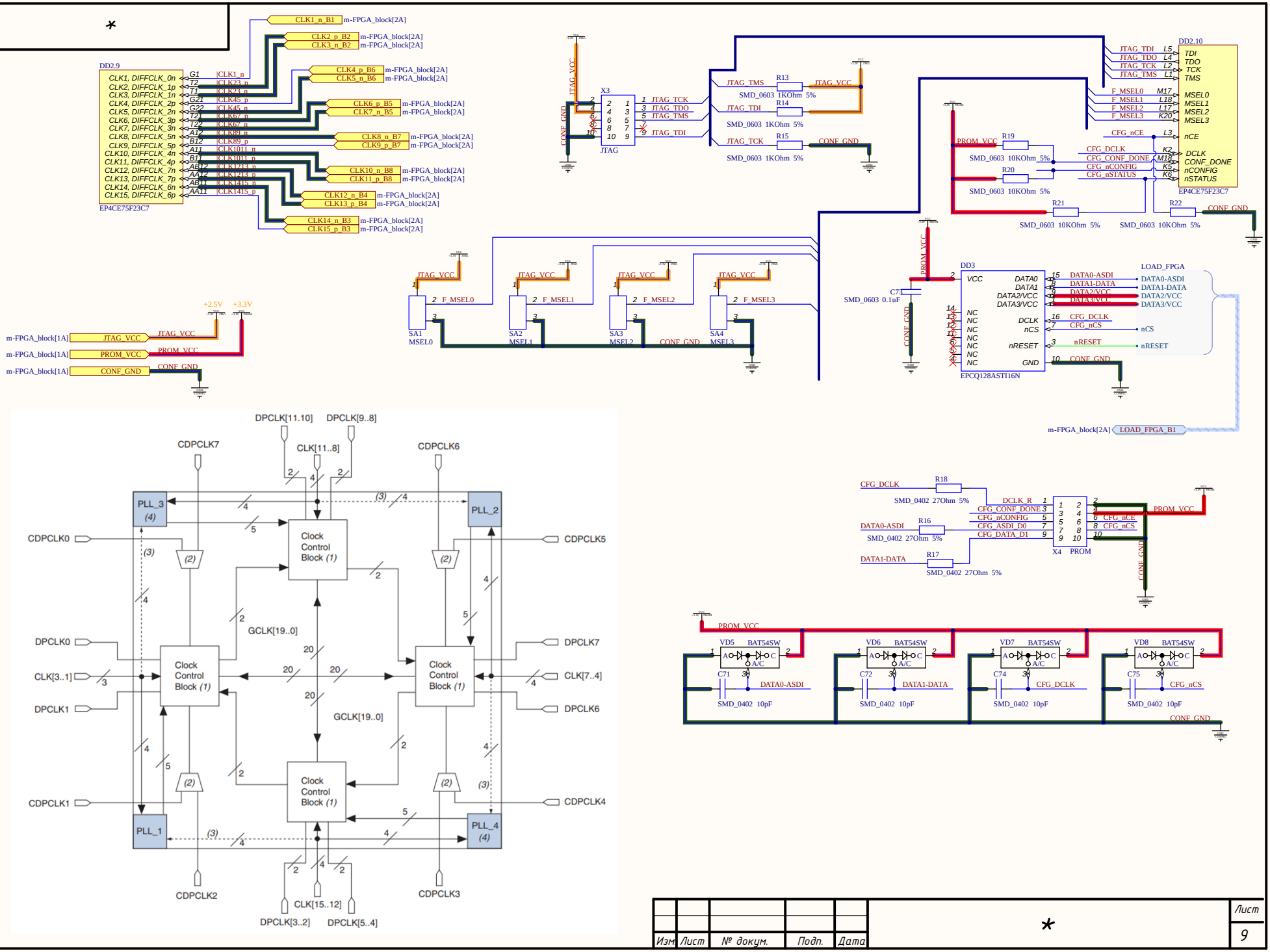

*
LOAD_FPGA_PS
m-FPGA_block[2A]
LOAD_FPGA
D0-ASDI
D1-DATA
F-nCS
DATA0-ASDI
DATA1-DATA
CFG_nCS
SDR2_CMD
m-FPGA_block[3A]
SDR2_CMD_B1
SDR2_WE
SDR2_CAS
SDR2_RAS
SDR2_CS
SDR2_BS0
SDR2_BS1
SDR2_CKE
DD2.1
BANK 1
EP4CE75F23C7
m-FPGA_block[2A]
LED1_B1
LED2_B1
USB_BE
USB_BE_B2
USB_BE[0..3]
USB_BE0
USB_BE1
USB_BE2
USB_BE3
USB_CMD
USB_CMD_B2
USB_WAKEUPn
USB_TXEn
USB_RXFn
USB_OEn
USB_RDn
USB_WRn
USB_SIWUn
USB_CONF_GPIO0
USB_CONF_GPIO1
DD2.2
BANK 2
USB_RST
FPGA_B12_USB_RST
FPGA_DUP1
FPGA_RDN1
R7
R8
SMD_0402 0Ohm 5%
VCC_B12
VSS_B12
Подп. и дата
Инв. № дубл.
Взам. инв. №
Подп. и дата
Инв. № подл.
Изм
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№ докум.
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Копировал
Формат А3
Форма 2а ГОСТ 2.104-2006


*
m-FPGA_block[4A]
USB_DATA_B3
USB_DATA
USB_FIFO_D[0..31]
USB_D[0..31]
DD2.3
BANK 3
EP4CE75F23C7
SRD_DQ
m-FPGA_block[4A]
SDR1_DATA_B4
1DQ[0...31]
DD2.4
BANK 4
EP4CE75F23C7
m-FPGA_block[4A]
SDR1_DQM0
SDR1_DQM2
m-FPGA_block[4A]
CLK_OUT_B4_SDR1
CLK_O_SDR1
VSS_B34
m-FPGA_block[4A]
VCC_B34
VCC_B34
m-FPGA_block[4A]
VSS_B34
VSS_B34
Подп. и дата
Инв. № дубл.
Взам. инв. №
Подп. и дата
Инв. № подл.
Изм
Лист
№ докум.
Подп.
Дата
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Форма 2а ГОСТ 2.104-2006
Копировал
Формат А3


*
DD2.5
BANK 5
1SDR_A4 AA21
FPGA_DUP3 T17
FPGA_RDN3 T18
1SDR_A8 W20
1SDR_CKE W19
1SDR_A1 Y22
1SDR_DQM1 Y21
1SDR_A7 U20
1SDR_CAS U19
1SDR_RAS W22
1SDR_A11 W21
1SDR_A3 T20
1SDR_A5 T19
R17
1SDR_CS V22
1SDR_WE V21
R20
1SDR_A10 U22
1SDR_A2 U21
1SDR_A0 R18
1SDR_DQM3 R19
1SDR_BS1 R22
1SDR_A6 R21
P22
1SDR_A9 P23
1SDR_BS0 P21
N20
N19
N18
LED4 N22
LED3 N21
M22
M21
M20
M19
M16
m-FPGA_block[4A] SDR1_CKE
m-FPGA_block[4A] SDR1_DQM1
m-FPGA_block[4A] SDR1_CAS
m-FPGA_block[4A] SDR1_RAS
m-FPGA_block[4A] SDR1_CS
m-FPGA_block[4A] SDR1_WE
m-FPGA_block[4A] SDR1_DQM3
m-FPGA_block[4A] SDR1_BS1
m-FPGA_block[4A] SDR1_BS0
m-FPGA_block[4A] LED4_B5
m-FPGA_block[4A] LED3_B5
IO, DIFFIO_R41p, (DM3R/BWS#3R)/(DM3R/BWS#3R)/(DM1R/BWS#1R)
IO, RUP3
IO, RDN3
IO, (DQS3R/CQ3R#,CDPCLK4)/(DQS3R/CQ3R#,CDPCLK4)/(DQS3R/CQ3R#,CDPCLK4)
IO, VREFB5N2
IO, DIFFIO_R40n, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R40p
IO, DIFFIO_R39n, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R39p
IO, DIFFIO_R37n, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R37p, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R36n
IO, DIFFIO_R36p
IO, DIFFIO_R35n
IO, DIFFIO_R34n, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R34p, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R33n, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R32n, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R32p, (DQ3R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R31n
IO, DIFFIO_R31p, (DM1R/BWS#1R)/(DM3R/BWS#3R)/(DM1R/BWS#1R)
IO, DIFFIO_R29n, (DQ1R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R29p, (DQ1R)/(DQ3R)/(DQ1R)
IO, VREFB5N1
IO, DIFFIO_R27n, (DQ1R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R27p, (DQ1R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R25n, (DQ1R)/(DQ3R)/(DQ1R)
IO, VREFB5N0
IO, DIFFIO_R24p, (DQS1R/CQ1R#,DPCLK6)/(DQS1R/CQ1R#,DPCLK6)/(DQS1R/CQ1R#,DPCLK6)
IO, DIFFIO_R23n, (DEV_OE)
IO, DIFFIO_R23p, (DEV_CLRn)
IO, DIFFIO_R22n, (DQ1R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R22p, (DQ1R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R21n, (DQ1R)/(DQ3R)/(DQ1R)
IO, DIFFIO_R21p, (DQ1R)/(DQ3R)/(DQ1R)
IO
EP4CE75F23C7
FPGA_DUP3
R46
SMD_0402 0Ohm 5%
VCC_B56
FPGA_RDN3
R47
SMD_0402 0Ohm 5%
VSS_B56
SRD_ADR
m-FPGA_block[3A] SDR1_ADR_B5
A0 1SDR_A0
A1 1SDR_A1
A2 1SDR_A2
A3 1SDR_A3
A4 1SDR_A4
A5 1SDR_A5
A6 1SDR_A6
A7 1SDR_A7
A8 1SDR_A8
A9 1SDR_A9
A10 1SDR_A10
A11 1SDR_A11
FC_CMD
m-FPGA_block[3A] FC_CMD_B6
F_BOE[0..1] FC_BUS_BOE[0..1]
FC_BUS_BOE0
FC_BUS_BOE1
F_TXPER FC_TXPER
F_OELE FC_OELE
F_RFEN FC_RFEN
F_BISTLE FC_BISTLE
F_RXLE FC_RXLE
F_LPEN FC_LPEN
F_LFI FC_LFI
F_TRSTZ FC_TRSTZ
FC_TX
m-FPGA_block[3A] FC_TX_B6
F_TXOP FC_TXOP
F_TXCT0 FC_TXCT0
F_TXCT1 FC_TXCT1
F_SCSEL FC_SCSEL
F_TX[0..7] FC_BUS_TX[0..7]
FC_BUS_TX0
FC_BUS_TX1
FC_BUS_TX2
FC_BUS_TX3
FC_BUS_TX4
FC_BUS_TX5
FC_BUS_TX6
FC_BUS_TX7
DD2.6
BANK 6
L22
L21
K19
K22
K21
J22
J21
H22
F_TRSTZ
FC_TXPER H21
K18
J18
F22
FC_TXOP
FC_SCSEL F21
FC_BUS_BOE0 J20
FC_LPEN J19
FC_BUS_BOE1 H20
FC_OELE H19
FC_BUS_TX0 E22
FC_BUS_TX3 E21
H18
FC_BUS_TX1 D22
FC_BUS_TX2 D21
FC_RXLE F20
FC_RFEN F19
FC_BISTLE G18
H17
FC_BUS_TX4 C22
FC_BUS_TX5 C21
FC_BUS_TX7 B22
FC_TXCT0 B21
FC_TXCT1 C20
FC_BUS_TX6 D20
FC_LFI F17
IO, DIFFIO_R20n, (INIT_DONE)
IO, DIFFIO_R20p, (CRC_ERROR)
IO, VREFB6N2
IO, DIFFIO_R19n, (nCEO)
IO, DIFFIO_R19p, (CLKUSR)
IO, DIFFIO_R18n, (DQS0R/CQ1R,DPCLK7)/(DQS0R/CQ1R,DPCLK7)/(DQS0R/CQ1R,DPCLK7)
IO, DIFFIO_R18p, (DM0R)/(DM1R/BWS#1R)/(DM1R/BWS#1R)
IO, DIFFIO_R17n, (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R17p, (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R16n, (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R16p
IO, DIFFIO_R14n, (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R14p, (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R13n
IO, DIFFIO_R13p
IO, DIFFIO_R12n, (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R12p, (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R11n, (nWE), (DQ0R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R11p, (nOE), (_)/(DQ1R)/(DQ1R)
IO, VREFB6N1
IO, DIFFIO_R10n, (DM2R)/(DM1R/BWS#1R)/(DM1R/BWS#1R)
IO, DIFFIO_R10p, (_)/(DQ1R)/(DQ1R)
IO, DIFFIO_R9n, (nAVD), (DQ2R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R9p, (DQ2R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R8n, (PADD23), (DQ2R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R8p
IO, DIFFIO_R6n, (DQ2R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R6p, (DQ2R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R5n, (PADD22), (DQ2R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R5p, (PADD21), (DQ2R)/(DQ1R)/(DQ1R)
IO, DIFFIO_R4n, (PADD20), (DQS2R/CQ3R,CDPCLK5)/(DQS2R/CQ3R,CDPCLK5)/(DQS2R/CQ3R,CDPCLK5)
IO, VREFB6N0
IO, DIFFIO_R1n, (DQ2R)/(DQ1R)/(DQ1R)
EP4CE75F23C7
m-FPGA_block[3A] VCC_B56
m-FPGA_block[3A] VSS_B56
Подп. и дата
Инв. № дубл.
Взам. инв. №
Подп. и дата
Инв. № подл.
Изм Лист № докум. Подп. Дата
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Лист
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Форма 2а ГОСТ 2.104-2006
Копировал
Формат А3


*
FC_RX
F_RXST[0..2]
FC_RXST[0..2]
FC_RXST2
FC_RXST1
FC_RXST0
F_RXOP
FC_RXOP
FC_BUS_RX[0..7]
FC_BUS_RX[0..7]
FC_BUS_RX0
FC_BUS_RX1
FC_BUS_RX2
FC_BUS_RX3
FC_BUS_RX4
FC_BUS_RX5
FC_BUS_RX6
FC_BUS_RX7
m-FPGA_block[4A]
FC_RX_B7
SRD_ADR
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
2SDR_A0
2SDR_A1
2SDR_A2
2SDR_A3
2SDR_A4
2SDR_A5
2SDR_A6
2SDR_A7
2SDR_A8
2SDR_A9
2SDR_A10
2SDR_A11
m-FPGA_block[4B]
2SDR_ADR_B7
m-FPGA_block[4A]
2SDR_DQM[0..3]
2SDR_DQM0
2SDR_DQM1
2SDR_DQM2
2SDR_DQM3
DD2.7
BANK 7
IO, DIFFIO_T46n, (DQ2T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T46p, (DQ2T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T45p, (DQS0T/CQ1T,CDPCLK6)/(DQS0T/CQ1T,CDPCLK6)/(DQS0T/CQ1T,CDPCLK6)
IO, DIFFIO_T44n
IO, DIFFIO_T44p
IO, VREFB7N0
IO, DIFFIO_T42n, (DQ2T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T42p, (DQ2T)/(DQ5T)/(DQ5T)
IO, PLL2_CLKOUTn
IO, PLL2_CLKOUTp
IO, DIFFIO_T40p, (DQ2T)/(DQ5T)/(DQ5T)
IO, RUP4
IO, RDN4
IO, DIFFIO_T39n, (DQ2T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T39p, (PADD0)
IO, DIFFIO_T38n
IO, DIFFIO_T38p, (DQ2T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T36n, (PADD1), (DQ2T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T36p, (PADD2), (_)/(DQ5T)/(DQ5T)
IO, DIFFIO_T35n, (DM2T)/(DM5T/BWS#5T)/(DM5T/BWS#5T)
IO, DIFFIO_T35p, (DQ4T)/(DQ5T)/(DQ5T)
IO, VREFB7N1
IO, DIFFIO_T33n, (PADD3), (DQ4T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T33p, (PADD4), (DQS2T/CQ3T,DPCLK8)/(DQS2T/CQ3T,DPCLK8)/(DQS2T/CQ3T,DPCLK8)
IO, DIFFIO_T32n, (PADD5), (DQ4T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T32p, (PADD6), (DQ4T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T31n, (PADD7)
IO, DIFFIO_T31p, (PADD8), (DQ4T)/(DQ5T)/(DQ5T)
IO, VREFB7N2
IO, DIFFIO_T27n, (PADD9), (DQ4T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T27p, (PADD10), (DQ4T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T26n, (PADD11), (DQ4T)/(DQ5T)/(DQ5T)
IO, DIFFIO_T26p, (PADD12), (DQS4T/CQ5T,DPCLK9)/(DQS4T/CQ5T,DPCLK9)/(DQS4T/CQ5T,DPCLK9)
IO, (_)/(DQ5T)/(DQ5T)
IO, DIFFIO_T25n, (PADD13)
IO, DIFFIO_T25p, (PADD14), (DM4T)/(DM5T/BWS#5T)/(DM5T/BWS#5T)
EP4CE75F23C7
2SDR_A5
2SDR_A6
2SDR_A8
FC_BUS_RX7
FC_BUS_RX4
FC_BUS_RX6
2SDR_A7
FC_BUS_RX5
FC_BUS_RX3
FC_BUS_RX0
FC_BUS_RX2
2SDR_A3
FC_RXOP
FC_BUS_RX1
FC_RXST2
FC_RXST0
2SDR_A1
2SDR_A4
2SDR_DQM3
2SDR_DQM0
FC_RXST1
2SDR_A0
2SDR_A2
2SDR_DQM1
2SDR_A11
2SDR_DQM2
2SDR_A9
2SDR_A10
SRD_DQ
DQ0
DQ1
DQ2
DQ3
DQ4
DQ5
DQ6
DQ7
DQ8
DQ9
DQ10
DQ11
DQ12
DQ13
DQ14
DQ15
DQ16
DQ17
DQ18
DQ19
DQ20
DQ21
DQ22
DQ23
DQ24
DQ25
DQ26
DQ27
DQ28
DQ29
DQ30
DQ31
2DQ0
2DQ1
2DQ2
2DQ3
2DQ4
2DQ5
2DQ6
2DQ7
2DQ8
2DQ9
2DQ10
2DQ11
2DQ12
2DQ13
2DQ14
2DQ15
2DQ16
2DQ17
2DQ18
2DQ19
2DQ20
2DQ21
2DQ22
2DQ23
2DQ24
2DQ25
2DQ26
2DQ27
2DQ28
2DQ29
2DQ30
2DQ31
m-FPGA_block[4B]
2SDR_DATA_B8
DD2.8
BANK 8
IO, DIFFIO_T24n, (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T23n, (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T23p, (PADD15)
IO, DIFFIO_T22n, (PADD16), (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T22p, (PADD17), (DQS5T/CQ5T#,DPCLK10)/(DQS5T/CQ5T#,DPCLK10)/(DQS5T/CQ5T#,DPCLK10)
IO, VREFB8N0
IO, DIFFIO_T19n, (DATA2), (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T19p, (DATA3), (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T18n, (PADD18), (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T18p, (DATA4), (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T17n, (PADD19), (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T17p, (DATA15), (DQ5T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T16n
IO, DIFFIO_T15n, (DATA14), (DQS3T/CQ3T#,DPCLK11)/(DQS3T/CQ3T#,DPCLK11)/(DQS3T/CQ3T#,DPCLK11)
IO, DIFFIO_T15p, (DATA13), (DM5T/BWS#5T)/(DM3T/BWS#3T)/(DM5T/BWS#5T)
IO, DIFFIO_T14n
IO, DIFFIO_T14p
IO, DIFFIO_T13p, (DATA5), (DQ3T)/(DQ3T)/(DQ5T)
IO, VREFB8N1
IO, DIFFIO_T12p, (DATA6), (DQ3T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T11n, (DATA7), (DQ3T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T11p
IO, DIFFIO_T10n, (DQ3T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T10p, (DATA8), (DQ3T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T9n, (DATA9), (DQ3T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T7n, (DATA10), (DQ3T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T7p, (DATA11), (DQ3T)/(DQ3T)/(DQ5T)
IO, VREFB8N2
IO, DIFFIO_T6n
IO, DIFFIO_T3n, (DQ3T)/(DQ3T)/(DQ5T)
IO, DIFFIO_T3p, (DATA12), (DQS1T/CQ1T#,CDPCLK7)/(DQS1T/CQ1T#,CDPCLK7)/(DQS1T/CQ1T#,CDPCLK7)
IO, DIFFIO_T2n, (DM3T/BWS#3T)/(DM3T/BWS#3T)/(DM5T/BWS#5T)
IO, DIFFIO_T2p
IO
IO, PLL3_CLKOUTn
IO, PLL3_CLKOUTp
EP4CE75F23C7
2DQ2
2DQ15
2DQ11
2DQ14
2DQ12
2DQ13
2DQ10
2DQ8
2DQ31
2DQ29
2DQ27
2DQ28
2DQ1
2DQ4
2DQ3
2DQ6
2DQ9
2DQ25
2DQ26
2DQ0
2DQ16
2DQ5
2DQ24
2DQ20
2DQ17
2DQ23
2DQ19
2DQ30
2DQ7
2DQ21
2DQ22
2DQ18
m-FPGA_block[4B]
2SDR_OUT_B8
2SDR_CLK_B8
VSS_B78
m-FPGA_block[4A]
VCC_B78
m-FPGA_block[4A]
VSS_B78
Подп. и дата
Инв. № дубл.
Взам. инв. №
Подп. и дата
Инв. № подл.
Изм
Лист
№ докум.
Подп.
Дата
*
Лист
7
Форма 2а ГОСТ 2.104-2006
Копировал
Формат А3

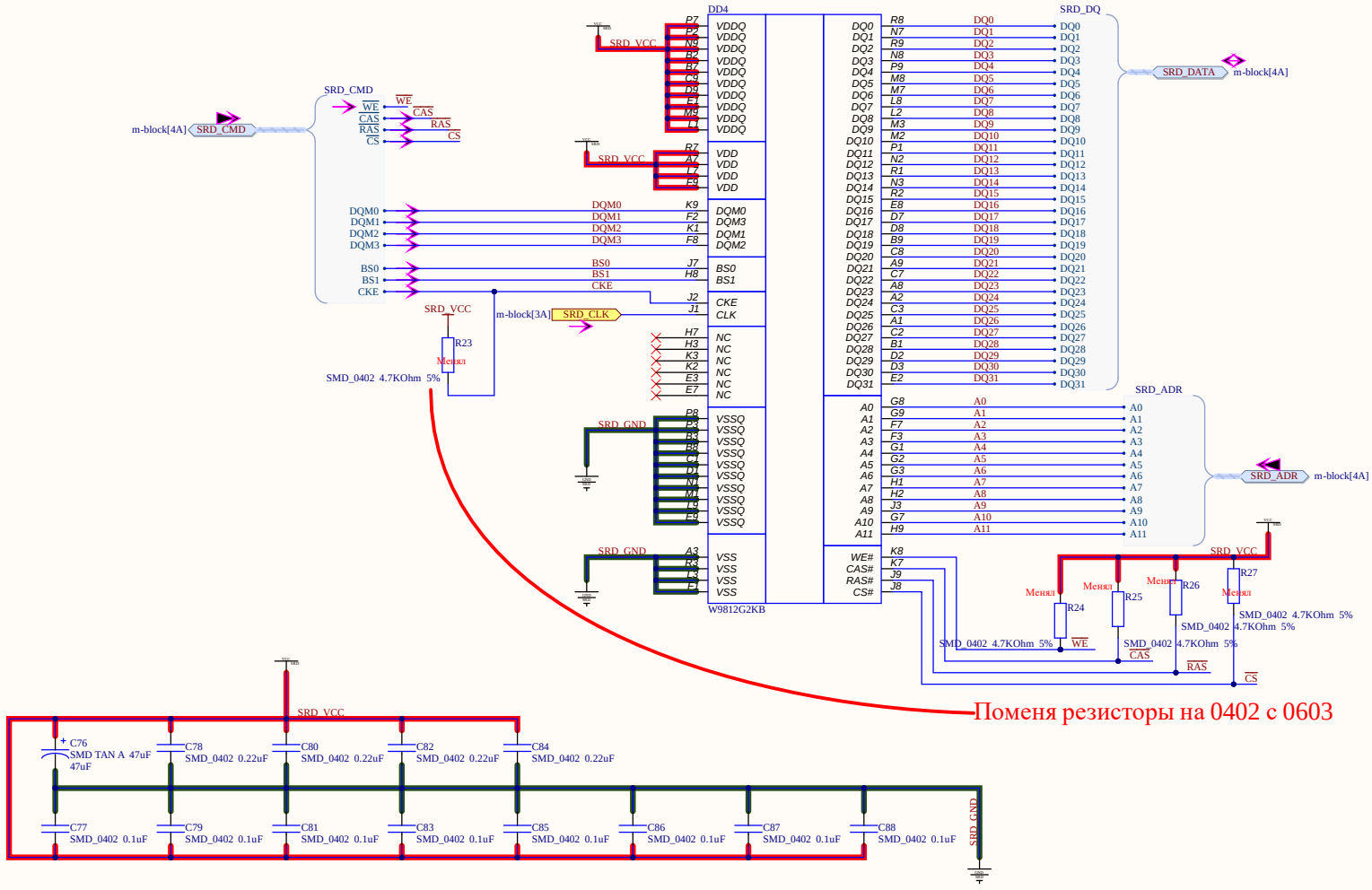

DD2.13
EP4CE75F23C7
DD2.11
EP4CE75F23C7
DD2.12
EP4CE75F23C7
F_VCCA
F_VCCPLL
F_AGND
F_VCCIO_B1
F_VCCIO_B2
F_VCCIO_B3
F_VCCIO_B4
F_VCCIO_B5
F_VCCIO_B6
F_VCCIO_B7
F_VCCIO_B8
F_VCC_CORE
F_GND
m-FPGA_block[2B]
SMD_0402 0.47uF
SMD_0402 0.22uF
SMD_0402 0.1uF
SMD TAN A 10uF / 10V
SMD TAN A 47uF
SMD_0603 4.7uF
Подп. и дата
Инв. № дубл.
Взам. инв. №
Подп. и дата
Инв. № подл.
Изм
Лист
№ докум.
Подп.
Дата
Лист
8
Форма 2а ГОСТ 2.104-2006
Копировал
Формат А3



Изм	Лист	№ докум.	Подп.	Дата	*	Лист 9

Подп. и дата | Инв. № дубл. | Взам. инв. № | Подп. и дата | Инв. № подл.

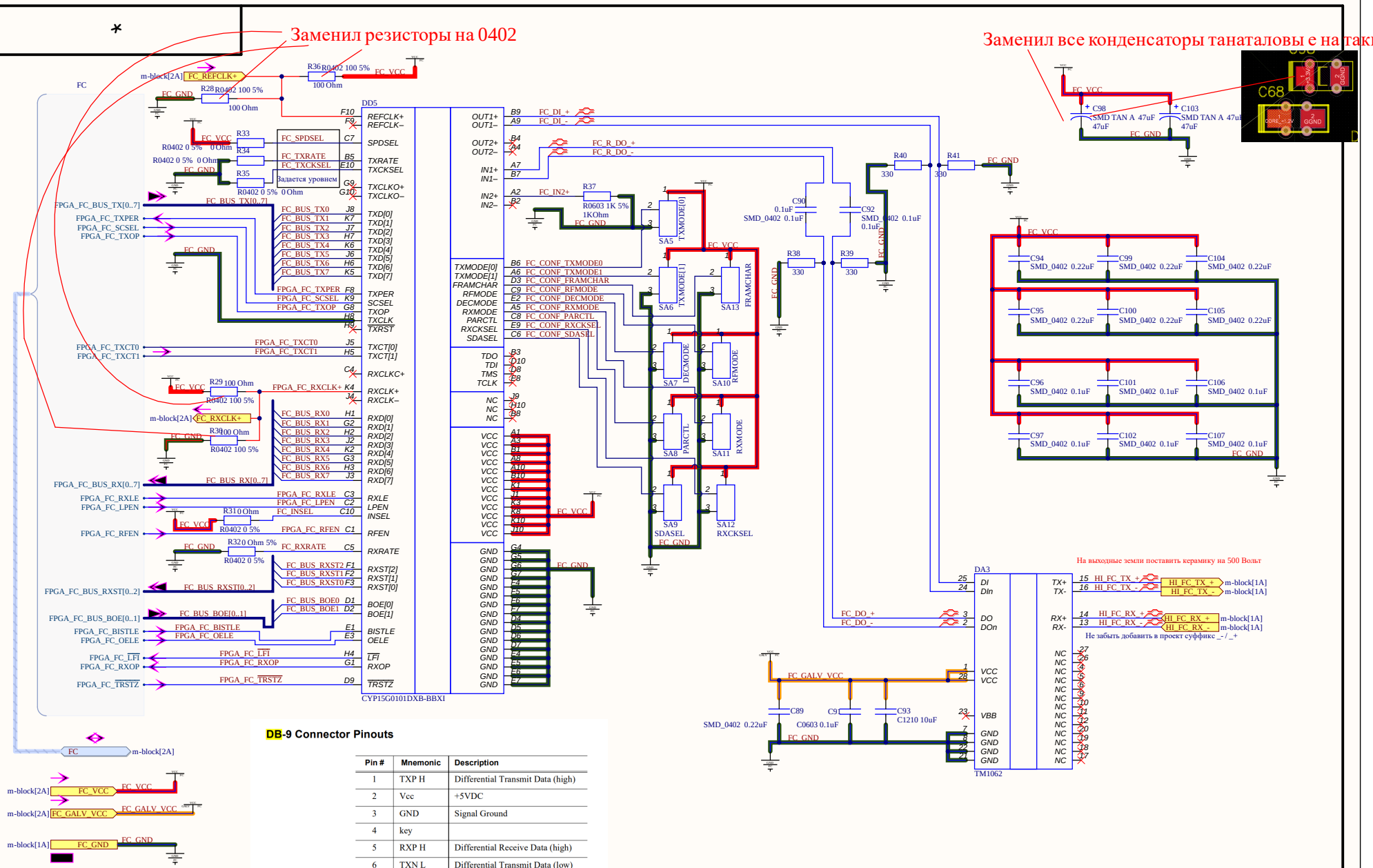
Форма 2а ГОСТ 2.104-2006 | Копировал | Формат А3



Подп. и дата | Инв. № дубл. | Взам. инв. № | Подп. и дата | Инв. № подл.

Изм	Лист	№ докум.	Подп.	Дата	*	Лист 10

Форма 2а ГОСТ 2.104-2006 | Копировал | Формат А3



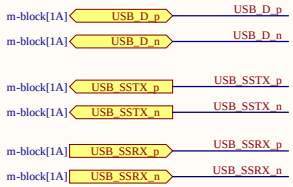
DB-9 Connector Pinouts

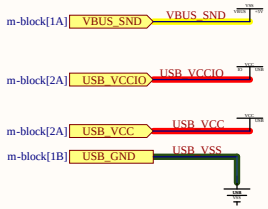
Pin #	Mnemonic	Description
1	TXP H	Differential Transmit Data (high)
2	Vcc	+5VDC
3	GND	Signal Ground
4	key	
5	RXP H	Differential Receive Data (high)
6	TXN L	Differential Transmit Data (low)
7	GND	Signal Ground
8	GND	Signal Ground
9	RXN L	Differential Receive Data (low)

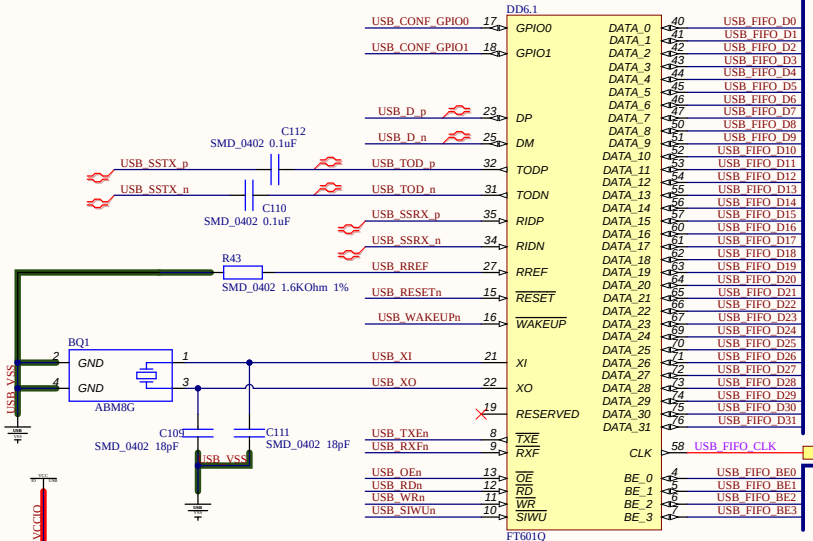
Подп. и дата | Инв. № дубл. | Взам. инв. № | Подп. и дата | Инв. № подл.

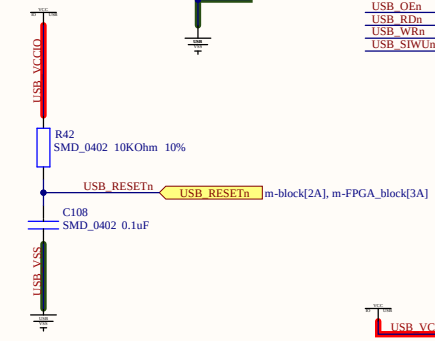
Изм	Лист	№ докум.	Подп.	Дата	*	Лист 11

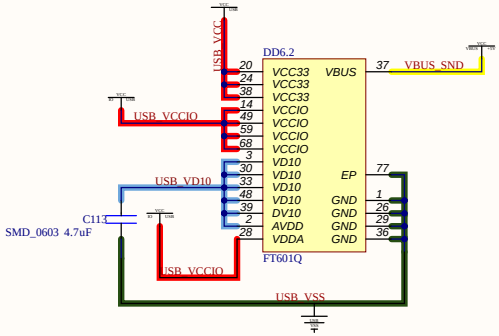
Форма 2а ГОСТ 2.104-2006 | Копировал | Формат А3

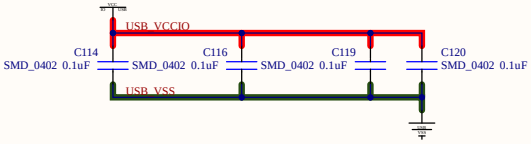


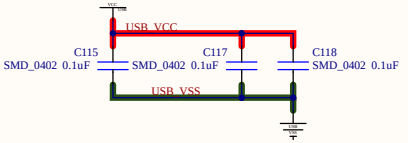


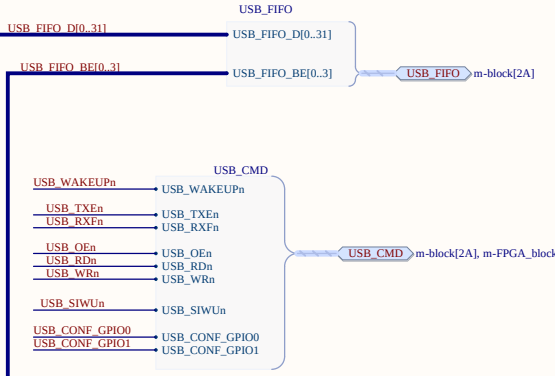










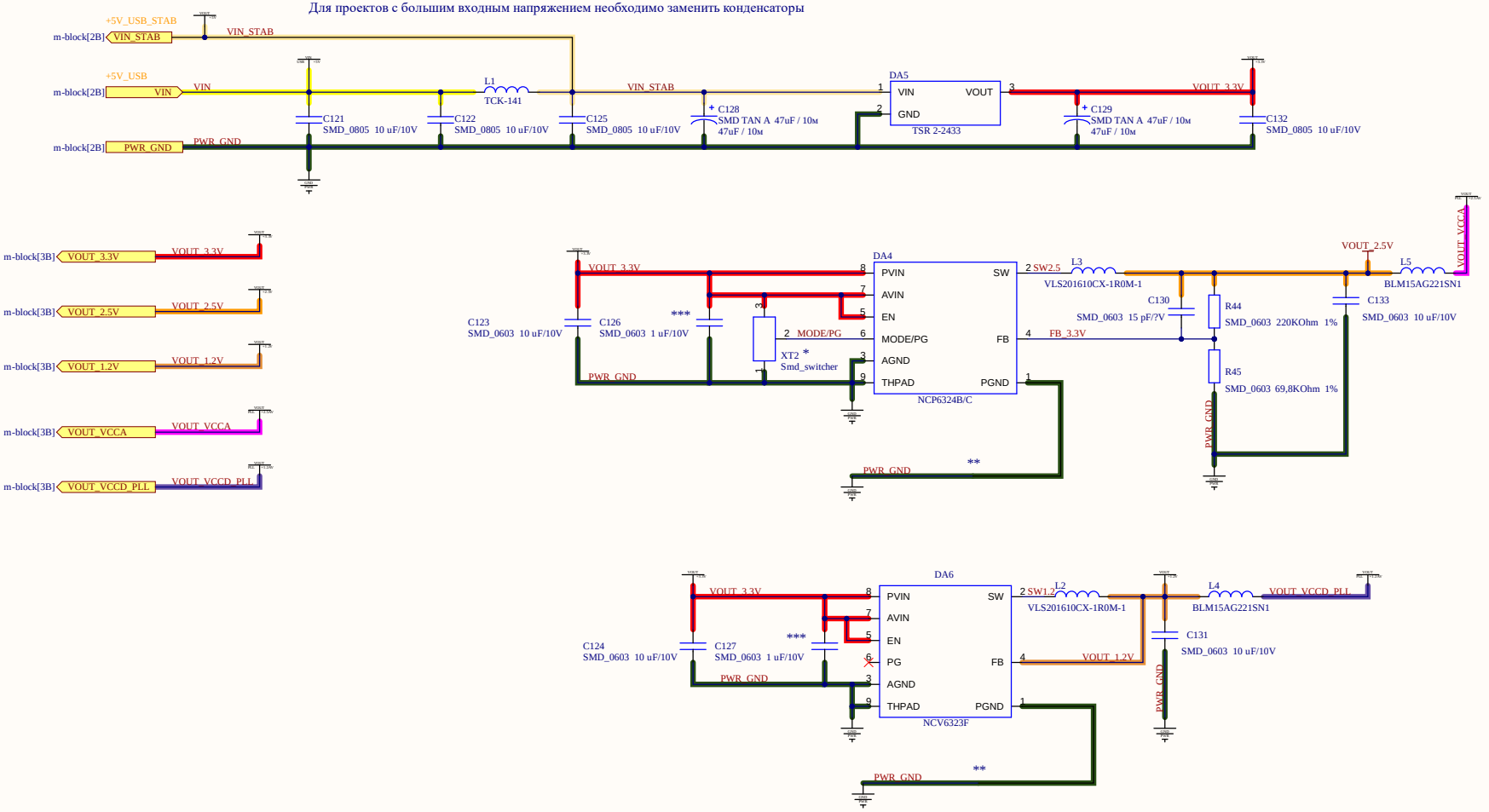


Изм	Лист	№ докум.	Подп.	Дата	*	Лист 12

Подп. и дата | Инв. № дубл. | Взам. инв. № | Подп. и дата | Инв. № подл.

*

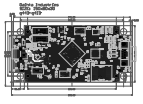
Для проектов с большим входным напряжением необходимо заменить конденсаторы



* Если применяется NCP6324C, то необходимо установить перемычку в зависимости от требунмой функции, если применяется NCP6324B перемчка не требуется
** PGDN1 и PGND2 должны иметь отдельные полигоны и соединяться с общей землёй в одной точке с помощью дорожки
*** Дополнительный конденсатор, который шунтирует вход AVIN на землю, типовой номинал 1uF. Устанавливается опционально.

Изм	Лист	№ докум.	Подп.	Дата	*	Лист 13

Подп. и дата | Инв. № дубл. | Взам. инв. № | Подп. и дата | Инв. № подл.



Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	ABM8G	ULTRA MINIATURE CERAMIC GLASS SEALED SMD CRYSTAL	BQ1	1							
	SMD_0603 0.1uF		C1, C73	2							
	C1210/0.027uF 500V		C2, C3	2							
	SMD_0402 0.1uF		C4, C5, C6, C17, C18, C19, C20, C30, C31, C36, C38, C47, C48, C49, C50, C51, C52, C55, C56, C57, C58, C63, C64, C65, C70, C77.1, C77.2, C79.1, C79.2, C81.1, C81.2, C83.1, C83.2, C85.1, C85.2, C86.1, C86.2, C87.1, C87.2, C88.1, C88.2, C90, C92, C96, C97, C101, C102, C106, C107, C108, C110, C112, C114, C115, C116, C117, C118, C119, C120	59							
	SMD_0402 0.47uF		C7, C8, C11, C12, C23, C24, C28, C33, C34, C35, C45, C53, C61, C66	14							
	SMD_0402 0.22uF		C9, C10, C13, C14, C15, C16, C21, C22, C25, C26, C27, C29, C32, C37, C39, C40, C41, C42, C43, C44, C46, C54, C62, C67, C78.1, C78.2, C80.1, C80.2, C82.1, C82.2, C84.1, C84.2, C89, C94, C95, C99, C100, C104, C105	38							
	SMD TAN A 10uF / 10V		C59	1							
	SMD_0603 4.7uF		C60, C65, C113	3							
	SMD TAN A 47uF		C68, C76.1, C76.2, C98, C103	5							
	SMD_0402 10pF		C71, C72, C74, C75	4							
	C0603/0.1uF		C91	1							
	C1210/ 10uF		C93	1							
	SMD_0402 18pF		C100, C111	2							
	SMD_0805 10 uF/ 10V		C121, C122, C125, C132	4							
	SMD_0603 10 uF/ 10V		C123, C124, C131, C133	4							
	SMD_0603 1 uF/ 10V		C126, C127	2							
	SMD TAN A 47uF / 6be		C128, C129	2							
	SMD_0603 15 pF/ 7V		C130	1							
	BVPAD		CN1, CN2, CN3, CN4, CN5, CN6, CN7	7							
	0.5_thPAD		CN8	1							
	FXOS6.000M3.35 M5	FXOS6.000M3.35 M5 FRONTER ELECTRONICS CO. LTD. China	DA1	1							
	106.25MHz	SM7745DEV-106.25M (https://www.dsi wafer.ru/product/SM7745DEV-106.25M)	DA2	1							
	TM1062		DA3	1							
	NCP632HB/C	Buck Converter-Synchronous, 3MHz, 2A	DA4	1							
	TSR 2-2433	TRACO POWER Non-isolated DC/DC Converter, TSR 2 Series, 2A	DA5	1							
	NCV6323F	Buck Converter-Synchronous, 3MHz, 1.6A	DA6	1							
	FDG1VC1102		DD1	1							
	EP4CE75F23C7	Cyclone IV Family FPGA, 2V Core, 292 I/O Pins, 4 PLLs, 484-Pin FBGA, Speed Grade 7, Commercial Grade	DD2	1				Digi-Key	EP4CE75F23C7-ND		
	EPCQ128ASTI16N		DD3	1							
	W9812G2KB	1Mx 4BANKS x 32 BITS SDRAM	DD4.1, DD4.2	2							
	CYP15G0101DXB-BBXI		DD5	1							
	FT601Q	32 Bit USB 3.0 to FIFO Bridge, 5 Gbps, 3 to 3.6 V, -40 to 85 degC, 76-Pin QFN, Pb-Free	DD6	1							
	TCK-141	TRACO POWER Inductor	L1	1							
	VLS252010CX-1R0M-1		L2, L3	2							
	BLM15AG221SN1		L4, L5	2							
	SMD_0402 00hm 5%		R1, R2, R5, R7, R8, R46, R47	7							
	SMD_0402 27Ohm 5%		R3, R4, R16, R17, R18	5							
	SMD_1210 0Ohm 5%		R6	1							
	SMD_0603 270Ohm 5%		R9, R10, R11, R12	4							
	SMD_0603 1kOhm 5%		R13, R14, R15	3							
	SMD_0603 10kOhm 5%		R19, R20, R21, R22	4							
	SMD_0402 4.7kOhm 5%		R23.1, R23.2, R24.1, R24.2, R25.1, R25.2, R26.1, R26.2, R27.1, R27.2	10							
	R0402 100 5%		R28, R29, R30, R36	4							
	R0402 0 5%		R31, R33, R34, R35	4							
	R0402 0 5%		R32	1							
	R0603 1K 5%		R37	1							
	SMD_0603 10kOhm 5%		R38, R39, R40, R41	4							
	SMD_0402 10KOhm 10%		R42	1							
	SMD_0402 1.8KOhm 1%		R43	1							
	SMD_0603 220KOhm 1%		R44	1							
	SMD_0603 69.8KOhm 1%		R45	1							
	MSEL0		SA1	1							
	MSEL1		SA2	1							
	MSEL2		SA3	1							
	MSEL3		SA4	1							
	TXMODE[0]		SA5	1							
	TXMODE[1]		SA6	1							
	SECMODE		SA7	1							
	FAIRCTL		SA8	1							
	SDASEL		SA9	1							
	RFMODE		SA10	1							
	SYMODE		SA11	1							
	RXCKSEL		SA12	1							
	FRAMCHAR		SA13	1							
	SMD_0805 - diod		VD1, VD2, VD3, VD4	4							
	BAT54SW		VD5, VD6, VD7, VD8	4							
	5310393-02		X1	1							
	DSUB-9		X2	1							
	JTAG		X3	1							
	PROM		X4	1							
	Smd_switcher		XT1, XT2	2							